

Verification Plan Report

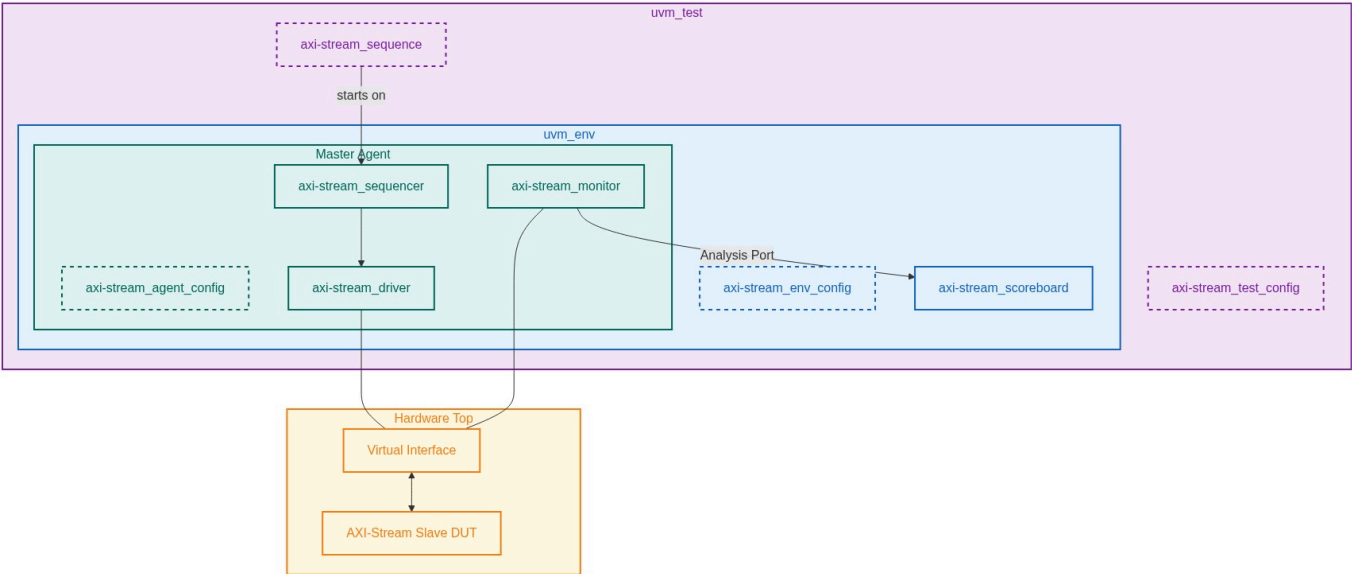
Project: AXI-Stream Slave DUT

Version: 3.0

1. Protocol Configurations

Feature	Status	Description
Ordering (OoO)	Strictly Ordered	Reordering is not permitted by protocol. All transfers must maintain absolute order.
Packet Structure	Supported (Aligned/Sparse)	Supports continuous aligned, unaligned, and sparse streams via TKEEP and TSTRB.
Interleaving	Optional (Transfer-level)	Interleaving is supported based on TID; can be disabled via Continuous_Packets property.
AXI5 Features	Supported	Includes TWAKEUP signaling and interface parity protection (Odd Parity).

2. UVM Environment Architecture



3. Test Requirements

Index	Name	Description
REQ_HANDSHAK E_01	TVALID/TREADY Handshake Rules	Verification of the 2-way flow control. TVALID must not wait for TREADY. TVALID must remain stable until TREADY is asserted.

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REQ_BYTE_QUAL_01	TKEEP and TSTRB Handling	Correct identification of valid bytes (TKEEP) and data vs position bytes (TSTRB).
REQ_PARITY_01	Odd Parity Protection	Identify single-bit flips on TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, and TUSER.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_RAND_PKT_SIZE	Send packets with randomized beat counts and TREADY backpressure.	REQ_HANDSHAKE_01
TC_SPARSE_STREAM	Verify slave can handle sparse streams with intermittent TKEEP/TSTRB bits.	REQ_BYTE_QUAL_01
TC_PARITY_ERR	Inject odd parity errors on TDATACHK and verify error signal detection.	REQ_PARITY_01

5. Functional Coverage Groups

Group Name	Description	TR Mapping
cg_handshake	Coverage for TVALID/TREADY being HIGH/LOW simultaneously.	REQ_HANDSHAKE_01
cg_data_width	Ensure all data bits in TDATA have toggled.	REQ_BYTE_QUAL_01

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	In	1	Clock
TVALID	In	1	Transmit valid
TREADY	Out	1	Receive ready
TDATA	In	DATA_WIDTH	Payload
TDATACHK	In	DATA_WIDTH/ 8	Parity bits (AXI5)

7. Verification Environment Structure

Root: axi_strm_slv_env/

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|-- agent : Master driving Slave
|-- env : UVM Environment
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